Q P Code:-25900508

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT202
ESSENTIALS OF APPLIED MATHEMATICS

(2024 ADMISSION ONWARDS)
Scheme For Valuation

Duration: 2 Hours Maximum Marks: 70

Part A

(Short Answer Questions)
Answer any 5 questions
Each question carries 2 marks

Question 1 Check whether $u(x, y) = 2xy$ is harmonic. **[K] / [C01]**

- Compute the partial derivatives

$$\frac{\partial u}{\partial x} = 2y \quad \Rightarrow \quad \frac{\partial^2 u}{\partial x^2} = 0$$
$$\frac{\partial u}{\partial y} = 2x \quad \Rightarrow \quad \frac{\partial^2 u}{\partial y^2} = 0$$

. . . 2 Marks

- Substitute into Laplace's equation

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

. . . 1 Mark

Total Marks should not exceed 2

Question 2 Find the derivative of $f(z) = z(1 - 4z)$. **[U] / [CO1]**

- $f'(z) = \frac{d}{dz}(z - 4z^2) = 1 - 8z$ . . . 2 Marks

Question 3 Find the particular solution of the differential equation $\frac{dy}{dx} = y$ that satisfies the initial condition $y = 5$ when $x = 0$ **[A] / [CO2]**

- For finding the general solution as $y = e^{x+c}$ . . . 2 Marks
- For Particular solution $y = 5e^x$ . . . 1 Mark

Question 4 Verify that the function $y = e^{3x}$ is a solution of the differential equation $y' - 3y = 0$. **[U] / [CO2]**

- Differentiate $y = e^{3x}$ with respect to x ... 1 Mark
- Substitute y and y' into the differential equation and verify ... 1 Mark

Question 5 Write the polar equation of a conic with one focus at the origin and the corresponding directrix $x = k, k > 0$ having eccentricity e **[K] / [CO3]**

- For correct polar equation $r = \dfrac{ek}{1 + e\cos\theta}$ ... 2 Marks

Question 6 Replace the polar equation $r^2 = 4r\sin\theta$ by equivalent Cartesian equation. . **[U] / [CO3]**

- Using $\begin{cases} x = r\cos\theta, \\ y = r\sin\theta, \\ r^2 = x^2 + y^2 \end{cases}$ ... 1 Mark
- Cartesian equation $\Rightarrow x^2 + y^2 = 4y$ ... 1 Mark

Question 7 Find the gradient of $f(x, y) = y - x$ at $(2, 1)$. **[U] / [CO4]**

- Compute $\dfrac{\partial f}{\partial x} = -1, \qquad \dfrac{\partial f}{\partial y} = 1$ ... 1 Mark
- $\nabla f(2, 1) = -\vec{i} + \vec{j}$ ... 1 Mark

Question 8 Define radius of curvature of a plane curve. **[K] / [CO4]**

- Correct definition ... 2 Marks

Part B
(Short Essay Questions)
Answer any 5 questions
Each question carries 6 marks

Question 9 Determine the value of the following limits: **[U] / [CO1]**

(a) $\lim\limits_{z\to -1}\left[\dfrac{z+2}{z+i}\right]$

- Answer is $\dfrac{-1+2}{-1+i} = \dfrac{1}{-1+i}$ ... 4 Marks

(b) $\lim\limits_{z\to 1}\left[\dfrac{iz^2-4}{z-1}\right]$

- Arguments and calculations to find the limit ... 2 Marks
- Limit does not exist ... 1 Mark

Total Marks should not Exceed 6

Question 10 Reduce $\dfrac{1+2i}{3-4i}+\dfrac{2-i}{5i}$ to a real number. **[U] / [CO1]**

- Simplify $\dfrac{1+2i}{3-4i}$ to-$\dfrac{1}{5}+\dfrac{2}{5}i$ ...3 Marks
- Simplify $\dfrac{2-i}{5i}$ to-$-\dfrac{1}{5}-\dfrac{2}{5}i$ ...3 Marks
- Final Answer $-\dfrac{2}{5}+0i$ ...1 Mark

Total Marks should not Exceed 6

Question 11 Given $z_1 = 2e^{i\pi/3}$ and $z_2 = 3e^{i\pi/6}$, find the value of the product $z_1 z_2$ in the form $a+ib$. **[U] / [CO1]**

- $z_1 z_2 = (2e^{i\pi/3})(3e^{i\pi/6}) = 6e^{i(\pi/3+\pi/6)} = 6e^{i\pi/2}$ ...4 Marks
- $6e^{i\pi/2} = 6i$ ...2 Marks

Question 12 Solve the differential equation $y' - y\tan x = 0$ using method of separation of variables.
. **[U] / [CO2]**

- For writing $\dfrac{dy}{y} = \tan x\, dx$ ...3 Marks
- $\displaystyle\int \frac{dy}{y} = \int \tan x\, dx \quad \Rightarrow \quad \ln|y| = -\ln|\cos x| + C$ ...3 Marks
- For writing $y = C\sec x$ ...1 Mark

Total Marks should not Exceed 6

Question 13 Find the equation of the hyperbola when $\dfrac{y^2}{4}-\dfrac{x^2}{5}=1$ is shifted 2 units down. Also find the center, foci, vertices and eccentricity of the new hyperbola.
. **[U] / [CO3]**

- Equation of the new hyperbola is $\dfrac{(y+2)^2}{4}-\dfrac{x^2}{5}=1$ ...2 Marks
- Calculations to find **Center:** $(0,-2)$ ...2 Marks
- Calculations to find **Vertices** $(0,0)$ and $(0,-4)$ ...2 Marks
- Calculations to find **Foci**:$(0,1)$ and $(0,-5)$. ...1 Mark
- Calculations to find **Eccentricity**:$\dfrac{3}{2}$ ...1 Mark

Total Marks should not Exceed 6

Question 14 Find the vertices, foci, eccentricity and directrices of the ellipse $6x^2+9y^2=54$ **[U] / [CO3]**

- For writing $6x^2+9y^2=54 \implies \dfrac{x^2}{9}+\dfrac{y^2}{6}=1$ ...2 Marks

- Calculations to find **Center:** (0,0) ... 1 Mark
- Calculations to find **Vertices**$(\pm 3, 0)$ ... 1 Mark
- Calculations to find **Foci**: $(\pm\sqrt{3}, 0)$ ... 1 Mark
- Calculations to find **Eccentricity**:$\dfrac{\sqrt{3}}{3}$ ... 1 Mark
- Calculations to find **Directrices:** $x = \pm 3\sqrt{3}$ ... 1 Mark

Total Marks should not Exceed 6

Question 15 The position vector of a particle at time t is given by **[A] / [CO4]**

$$\vec{r}(t) = \sin t\hat{i} + (t^2 - \cos t)\hat{j} + e^t\hat{k}.$$

Find the particle's velocity, speed, direction of motion and acceleration at $t = 0$.

- Calculations to find **Velocity:** $(\cos t)\,\hat{i} + (2t + \sin t)\,\hat{j} + e^t\,\hat{k}$ ... 1 Mark
- Velocity at 0: $\hat{i} + 0 \cdot \hat{j} + \hat{k}$ ... 1 Mark
- **Speed at** $t = 0$: $|\vec{v}(0)| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$.] ... 1 Mark
- **Direction of motion at** $t = 0$:$\dfrac{1}{\sqrt{2}}\hat{i} + 0 \cdot \hat{j} + \dfrac{1}{\sqrt{2}}\hat{k}$ ... 2 Marks
- **Acceleration:** $\vec{a}(t) = (-\sin t)\,\hat{i} + (2 + \cos t)\,\hat{j} + e^t\,\hat{k}$ ... 1 Mark
- **Acceleration at** $t = 0$**:** $\vec{a}(0) = \langle -\sin 0,\ 2 + \cos 0,\ e^0\rangle$ ... 1 Mark

Total Marks should not Exceed 6

Question 16 Find the principal unit normal vector to the curve **[U] / [CO4]**

$$\mathbf{r}(t) = (1 - 2\sin t)\hat{i} + (3 - 2\cos t)\hat{j}.$$

- Velocity (tangent) vector $\mathbf{v}(t) = \dfrac{d\mathbf{r}}{dt} = (-2\cos t)\hat{i} + (2\sin t)\hat{j}$ ... 2 Marks
- Unit tangent vector $\mathbf{T}(t) = \dfrac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = -\cos t\,\hat{i} + \sin t\,\hat{j}$ ... 2 Marks
- Principal unit normal vector

$$\mathbf{N}(t) = \frac{\mathbf{T}(t)}{|\mathbf{T}(t)|} = \sin t\,\hat{i} + \cos t\,\hat{j}$$

... 3 Marks

Total Marks should not Exceed 6

Part C
(Essay Questions)
Answer any 3 questions
Each question carries 10 marks

Question 17 (a) Find the real and imaginary parts of the function $f(z) = z^2$ both in Cartesian co-ordinates and polar co-ordinates. **[U] / [CO1]**

- **Cartesian Form of** $f(z) = z^2$
  - Take $z = x + iy$ in $f(z)$ ... 2 Marks
  - $f(z) = (x+iy)^2 = x^2 + 2ixy + i^2y^2 = (x^2 - y^2) + i(2xy)$ ... 2 Marks
- **Polar Form** $f(z) = z^2$
  - Take $z = re^{i\theta}$ ... 2 Marks
  - $f(z) = f(z) = (re^{i\theta})^2 = r^2e^{i2\theta} = r^2(\cos 2\theta + i \sin 2\theta)$ ... 2 Marks

(b) Express the function $f(z) = \dfrac{1}{z}$ in the form $f(z) = u(x,y) + iv(x,y)$.

- $u(x,y) + iv(x,y)$ **form of** $f(z) = \dfrac{1}{z}$ **.**
  - Writing $f(z) = \dfrac{1}{x+iy} = \dfrac{x-iy}{x^2+y^2}$ ... 4 Marks
  - For getting $f(z) = \dfrac{x}{x^2+y^2} - i\dfrac{y}{x^2+y^2} = u(x,y) + iv(x,y)$ ... 1 Mark

Total Marks should not Exceed 10

Question 18 Show that the differential equation $\dfrac{x}{2}\cot y \dfrac{dy}{dx} = -1$ is exact and solve it.
. **[U] / [CO2]**

- For writing

$$\frac{x}{2}\cot y \, \frac{dy}{dx} = -1 \qquad \Longrightarrow \qquad x \cot y \, dy + 2\, dx = 0.$$

... 3 Marks

- Getting $M(x,y) = 2, \qquad N(x,y) = x\cot y.$ ... 2 Marks
- For calculating

$$\frac{\partial M}{\partial y} = 0, \qquad \frac{\partial N}{\partial x} = \cot y,$$

... 3 Marks

- For writing the conclusion that the equation is not exact ... 2 Marks

Question 19 (a) Solve the differential equation $y' - 2xy = xe^{x^2}$. **[U] / [CO2]**

- Taking $P = -2x$ ... 1 Mark
- For finding the integrating factor $e^{\int P\, dx} = e^{-x^2}$ ... 3 Marks

- Rest of the calculations and answer $y = e^{x^2}\left(\frac{x^2}{2} + C\right)$ ... 3 Marks

(b) Convert the differential equation $xy' - y = x^3$ into standard linear form and solve it.

- Divide by x : $y' - \frac{1}{x}y = x^2$. ... 2 Marks
- Taking $P(x) = -\frac{1}{x}$ and calculating integrating factor $e^{\int P\,dx} = e^{-\ln|x|} = \frac{1}{|x|}$ . ... 2 Marks
- Rest of the calculations and answer $y = x\left(\frac{x^2}{2} + C\right) = \frac{x^3}{2} + Cx$ ... 3 Marks

Total Marks should not Exceed 10

Question 20 Find the equation of the new parabola, when the parabola $y^2 = 8x$ is shifted vertically down by 2 units and horizontally to the right by 1 unit. Also find the focus, vertex, directrix and focal axis of the new parabola. **[U] / [CO3]**

- Shifting the curve right by 1 and down by 2 means replace

$$x \mapsto x - 1, \qquad y \mapsto y + 2.$$

... 2 Marks

- New parabola is

$$(y + 2)^2 = 8(x - 1).$$

... 2 Marks

- Calculations to find **Vertex:** (1,-2) ... 2 Marks
- Calculations to find **Focus:** (3,-2) ... 2 Marks
- Calculations to find **Directrix:** x=-1 ... 2 Marks
- Calculations to find **Focal axis:** y=-2 ... 2 Marks

Total Marks should not Exceed 10

Question 21 (a) Find the length of the curve **[U] / [CO4]**

$$\vec{r}(t) = (1 - t)\hat{i} + 2t\hat{j} + 3t\hat{k}, \quad 0 \le t \le 1.$$

- Differentiate:

$$\vec{r}\,'(t) = -\hat{i} + 2\hat{j} + 3\hat{k}.$$

... 2 Marks

- For finding

$$|\vec{r}\,'(t)| = \sqrt{(-1)^2 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}.$$

... 2 Marks

- Length of the curve:

$$L = \int_0^1 |\vec{r}\,'(t)|\, dt = \int_0^1 \sqrt{14}\, dt = \sqrt{14}.$$

... 2 Marks

(b) Find the arc length parametrization of the curve

$$\vec{r}(t) = \cos t\hat{i} + \sin t\hat{j} + t\hat{k}$$

with base point $t_0 = 0$.

- For finding $\vec{r}\,'(t) = -\sin t\,\hat{i} + \cos t\,\hat{j} + 1\cdot\hat{k}$ ... 2 Marks
- Arc length$= \int_0^t |\vec{r}\,'(u)|\, du = \int_0^t \sqrt{2}\, du = \sqrt{2}\, t.$ ... 2 Marks
- For writing arc length parametrization ... 2 Marks

Total Marks should not Exceed 10

Question 22 If $\vec{r}(t) = 2\cos t\hat{i} - \sin t\hat{j} + 4t^2\hat{k}$, then find the values of **[U] / [CO4]**

(a) $\lim_{t\to 0} \vec{r}(t)$

- $\lim_{t\to 0} 2\cos t = 2, \qquad \lim_{t\to 0} -\sin t = 0, \qquad \lim_{t\to 0} 4t^2 = 0$
  .

  ... 2 Marks

- $\lim_{t\to 0} \vec{r}(t) = 2i$ ... 2 Marks

(b) $\lim_{t\to \frac{\pi}{2}} \vec{r}(t)$

- $\cos\frac{\pi}{2} = 0, \qquad -\sin\frac{\pi}{2} = -1, \qquad 4\left(\frac{\pi}{2}\right)^2 = \pi^2$ ... 2 Marks
- $\lim_{t\to \frac{\pi}{2}} \vec{r}(t) = -j + \pi^2 k$ ... 2 Marks

(c) $\frac{d\vec{r}}{dt}$ at $t = \frac{\pi}{2}$.

- $\frac{d\vec{r}}{dt} = -2\sin t\,\hat{i} - \cos t\,\hat{j} + 8t\,\hat{k}$ ... 2 Marks
- At $t = \frac{\pi}{2}$: $-2\sin\frac{\pi}{2} = -2,\ -\cos\frac{\pi}{2} = 0,\ 8\cdot\frac{\pi}{2} = 4\pi.$

$$\left.\frac{d\vec{r}}{dt}\right|_{t=\pi/2} = -2\vec{i} + 0\vec{j} + 4\vec{k}$$

... 2 Marks

Total Marks should not Exceed 10

N.B: Suitable stage marks may be fixed by the examiner for alternate correct answers.